

3.3 Statements and Functions

Statements

Each instruction we give the computer is a **statement**.

```
greetPerson("Ada");  
greetPerson("Grace");  
greetPerson("Alan");
```

Three statements, executed top to bottom.

A function

A named, reusable block of statements.

```
void greetPerson(std::string name) {  
    std::println("Hello, {}!", name);  
}
```

Call it as many times as you like, with different input each time.

Putting it together

```
#include <print>
#include <string>

void greetPerson(std::string name) {
    std::println("Hello, {}", name);
}

int main() {
    greetPerson("Ada");
    greetPerson("Grace");
    greetPerson("Alan");

    return 0;
}
```

Two kinds of errors

Compile-time error - the program won't even build.

```
greetPersonn("Ada"); // typo: no such function
```

Runtime error - it builds, but breaks while running.

```
int trouble = numerator / 0; // compiles fine, crashes when run
```

Next: 3.4

Talking to the User